



Effect of iron content on wear behavior of $\text{AlCoCrFe}_x\text{Mo}_{0.5}\text{Ni}$ high-entropy alloys

Chin-You Hsu^a, Tsing-Shien Sheu^b, Jien-Wei Yeh^{a,*}, Swe-Kai Chen^c

^a Department of Materials Science and Engineering, National Tsing Hua University, Hsinchu 300, Taiwan

^b Department of Air Transportation, Kainan University, Luzhu, Taoyuan 338, Taiwan

^c Center for Nanotechnology, Materials Science, and Microsystems, National Tsing Hua University, Hsinchu 300, Taiwan

ARTICLE INFO

Article history:

Received 18 October 2008

Received in revised form 28 October 2009

Accepted 30 October 2009

Available online 10 November 2009

Keywords:

Sliding wear

Hardness

Non-ferrous metals

Surface topography

ABSTRACT

A new high-entropy alloy system, AlCoCrFeMoNi , was designed based on the AlCoCrCuFeNi , by replacing Cu with Mo to improve strength and thermal stability. The effect of iron content on hardness and wear behavior of $\text{AlCoCrFe}_x\text{Mo}_{0.5}\text{Ni}$ alloys is related to the microstructural change. As the iron content is increased, the microstructure changes from “dendritic” for Fe0.6 and Fe1.0 alloys to “polygrain” for Fe1.5 and Fe2.0 alloys. The two constituent solution phases are BCC and σ , but the volume fraction of the BCC phase increases with the iron content. The hardness declines correspondingly because the σ phase is harder than the BCC phase. The wear resistance also declines as the Fe content increases, but the wear resistance of Fe2.0 is much lower than that of Fe1.5, which result is inconsistent with their similar hardness levels. Composition analysis of the worn surface and worn debris indicates that, the major wear mechanism of all these HE alloys is abrasion. The oxidation test at the pin/disk interface flash temperature, 500 °C, indicates that the oxidation rate of Fe2.0 markedly exceeds that of Fe1.5, indicating more oxides abrade the surface resulting lower wear resistance.

© 2009 Elsevier B.V. All rights reserved.

1. Introduction

For hundreds of years, alloy design has been mostly limited to a major metal element, such as ferrous, aluminum, copper, titanium, or magnesium, with the addition of minor elements to modify properties. A novel concept in alloy design for generating high-entropy alloys (HEAs) was proposed in 1995. HEAs by definition contain five or more major metal elements, each at over 5 at.% but less than 35 at.% [1]. Various alloy systems, such as Al-Co-Cr-Cu-Fe-Ni , $\text{Al-Co-Cr-Cu-Fe-Ni-B}$, $\text{Al-Co-Cr-Cu-Fe-Ni-Ti}$, $\text{Al-Co-Cr-Cu-Fe-Ni-V}$, Co-Cr-Cu-Fe-Ni-Ti , and Al-Co-Cr-Fe-Ni-Ti , reportedly exhibit promising properties, such as high strength, good oxidation and corrosion resistance, superior temper-softening resistance, and excellent abrasive wear resistance [2–8].

Among the reported systems, the microstructure and mechanical properties of $\text{Al}_x\text{CoCrCuFeNi}$ ($x=0\text{--}3.0$) alloys have been extensively investigated [3,9,10]. The microstructure is characterized by the existence of an interdendrite Cu-rich phase because elemental copper has a positive mixing enthalpy with the other elements in this alloy system and tends to be rejected by other elements in the dendrite. Therefore, earlier incipient melting at

around 1000 °C degrades temperature endurance. The soft Cu-rich region also lowers the wear resistance [10] and serves as a local galvanic local cell which promotes corrosion [11]. To overcome these shortcomings, this study develops a new $\text{AlCoCrFe}_x\text{Mo}_{0.5}\text{Ni}$ high-entropy alloy system. Mo is intended to replace Cu for avoiding the Cu segregation along interdendritic regions and promoting the strength at high temperatures because of its high melting point and elastic modulus. To get a significant effect without increasing too much cost from Mo, the molar amount of Mo is set to be one half of Al, Co, Cr, and Ni. Additionally, the effect of Fe content on the hardness and wear behavior of $\text{AlCoCrFe}_x\text{Mo}_{0.5}\text{Ni}$ high-entropy alloy is investigated and related to the microstructural change. The optimal amount of cheap Fe to be added is also evaluated.

2. Experimental methods

$\text{AlCoCrFe}_x\text{Mo}_{0.5}\text{Ni}$ high-entropy alloys with various iron contents ($x=0.6\text{--}2.0$ in molar ratio) were prepared by the arc smelting and casting method in a protective atmosphere of argon. The raw materials were granular elemental Al, Co, Cr, Fe and Ni with a purity of over 99 wt% and Fe–61 wt% Mo master alloy. The microstructure and chemical composition of the alloys were analyzed using scanning electron microscope (SEM, JEOL JSM-5410) and SEM X-ray energy dispersive spectrometry (EDS). Phases were identified using an X-ray diffractometer (XRD, Rigaku ME510-FM2) at a scanning speed of 4°/min and a scanning range from 20° to 100° using

* Corresponding author at: Department of Materials Science and Engineering, National Tsing Hua University, Room 315, Materials Science & Technology Building, Hsinchu 300, Taiwan. Tel.: +886 3 5742621x33868; fax: +886 3 5722366.

E-mail address: jwyeh@mx.nthu.edu.tw (J.-W. Yeh).

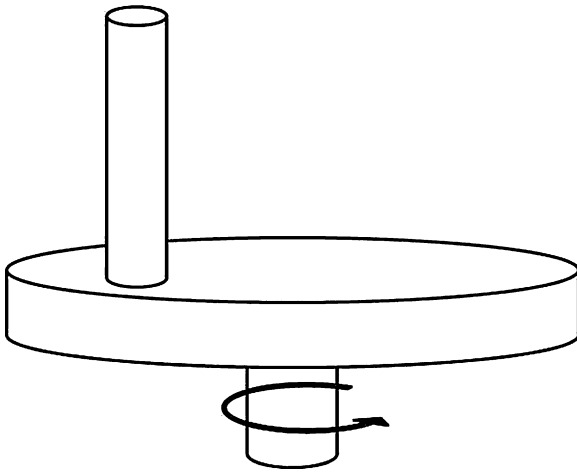


Fig. 1. Schematic wear test.

a copper target and an applied voltage and current of 30 kV and 20 mA, respectively. Macrohardness and microhardness measurements were made using Mitutoyo HM-115 and HV-115 hardness testing machines with loads of 5 kg and 10 g, respectively, with a duration in each case of 15 s. Wear behavior was examined using a pin-on-disk wear-testing machine under the dry sliding condition. The wear specimens were cylinders with a diameter of 8 mm and a height of 3 mm. Each specimen was fixed in a holder to wear against a 75-cm-diameter SKH51 steel disk with a hardness value of Hv 890, as shown in Fig. 1. The distance of the specimen from the center of the disk was 20 mm. The sliding velocity and time were 0.5 m/s and 24 h, respectively, with a normal load of 29.4 N. The wear resistance is defined as follows:

$$Wr = \frac{L}{\Delta V}$$

where Wr is the wear resistance, and ΔV is the volumetric loss of the specimen after it has slid through a distance L , and can be obtained by dividing the weight loss by its density. The worn surface and worn debris were also investigated using SEM and EDS. Furthermore, in order to investigate the oxidation effect on wear behavior oxidation test was conducted at 500 °C for 24 h in a box furnace to examine the oxidation rates of Fe1.5 and Fe2.0 alloys. The oxidizing temperature of 500 °C is chosen because the flash temperature at the pin/disk interface is calculated to be around this temperature [12].

3. Results and discussion

3.1. Constituent phase and microstructure

Fig. 2 presents the XRD patterns of AlCoCrFe_xMo_{0.5}Ni as-cast alloys with various Fe contents. The crystal structure is identified as a duplex BCC ($a = 0.287$ nm)–BCT ($a = 0.893$ nm and $c = 0.467$ nm), meaning that the constituent phases are a BCC solid solution phase and a σ phase. As the iron content increases, the peak intensity of BCC increases relative to that of σ , indicating that the amount of σ phase declines whereas that of the BCC phase increases. A comparison with the phases of AlCoCrCuFeNi alloy, on which this alloy is derived, indicates that FCC solid solution and Cu-rich phases no longer exist but are replaced by the σ phase in this alloy. The structure of the σ phase is similar to that of the CoCr phase, which is tetragonal with lattice constants of $a = 0.885$ nm and $c = 0.459$ nm. Similar structures, with slightly different lattice constants, are also observed in σ -FeCr and σ -FeMo phases, according to the JCPDS card. Related binary and ternary phase diagrams [13] show that the σ

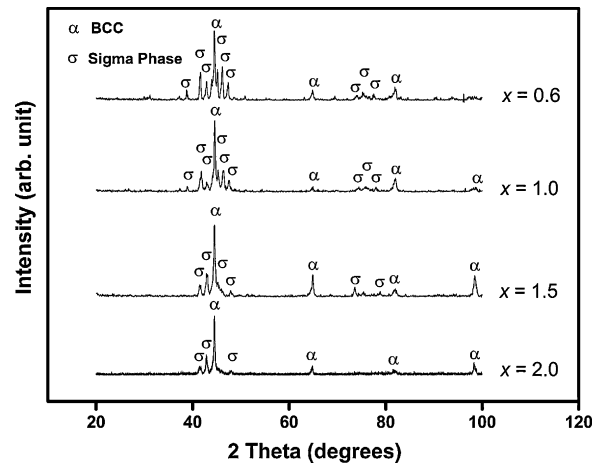


Fig. 2. XRD patterns of AlCoCrFe_xMo_{0.5}Ni alloys with various Fe contents.

phase effectively dissolves Fe and Ni. This result indicates that Mo in place of Cu in AlCoCrCuFeNi alloy promotes the formation of the σ phase.

Fig. 3 shows the as-cast microstructure of AlCoCrFe_xMo_{0.5}Ni alloys with various Fe contents. Fe0.6 and Fe1.0 alloys have typical cast dendritic structures while Fe1.5 and Fe2.0 alloys have poly-grain structures. The dendrites, interdendrites and grains further decompose into two fine phases. The dendrites contain a small amount of white phase but the interdendrites contain mostly white phase. The grains in Fe1.5 and Fe2.0 alloys also contain mostly white phase, which delineate the grain boundaries. Table 1 presents the compositions of the various regions and phases. Dendrites in the Fe0.6 and Fe1.0 alloys are richer in Al and Ni than is the overall alloy and the interdendrites are richer in Cr and Mo. The dark phase contains a higher concentration of Al and Ni and a lower concentration of Cr and Mo than the white phase. The above XRD analysis explains that, the dendrites have the BCC phase, derived from the binary NiAl compound, according to the pseudo-binary phase diagram of (Ni, Co, Fe)–(Al, Cr, Mo), further precipitating out the white σ phase during the cooling process after solidification. However, since the X-ray patterns do not show the ordered BCC peak at around 31°, the compound can be effectively regarded as a multi-element solid solution with at most a small degree of long-range order. Similarly, the Cr- and Mo-rich interdendrites, being the outer envelope of the cored dendrites, are also BCC phase

Table 1

Chemical compositions of AlCoCrFe_xMo_{0.5}Ni alloys in atomic percentage.

x-Value	Al	Cr	Fe	Co	Ni	Mo
0.6						
Nominal	19.6	19.6	11.8	19.6	19.6	9.8
DR	31.8	10.2	8.3	18.3	27.9	3.5
ID	14.4	23.8	13.5	19.9	14.8	13.6
1.0						
Nominal	18.2	18.2	18.2	18.2	18.2	9.1
DR	24.5	12.3	15.6	18.1	24.4	5.2
ID	11.3	21.0	19.3	17.5	14.1	16.9
1.5						
Nominal	16.7	16.7	25	16.7	16.7	8.3
Grain	16.9	16.5	25.6	16.0	16.9	7.9
Grain boundary	14.9	17.6	24.4	16.3	15.0	11.8
2.0						
Nominal	15.4	15.4	30.8	15.4	15.4	7.7
Grain	17.4	15.5	29.6	13.5	15.7	8.4
Grain boundary	16.8	14.7	27.9	14.9	14.1	11.7

DR: dendrite; IR: interdendrite.

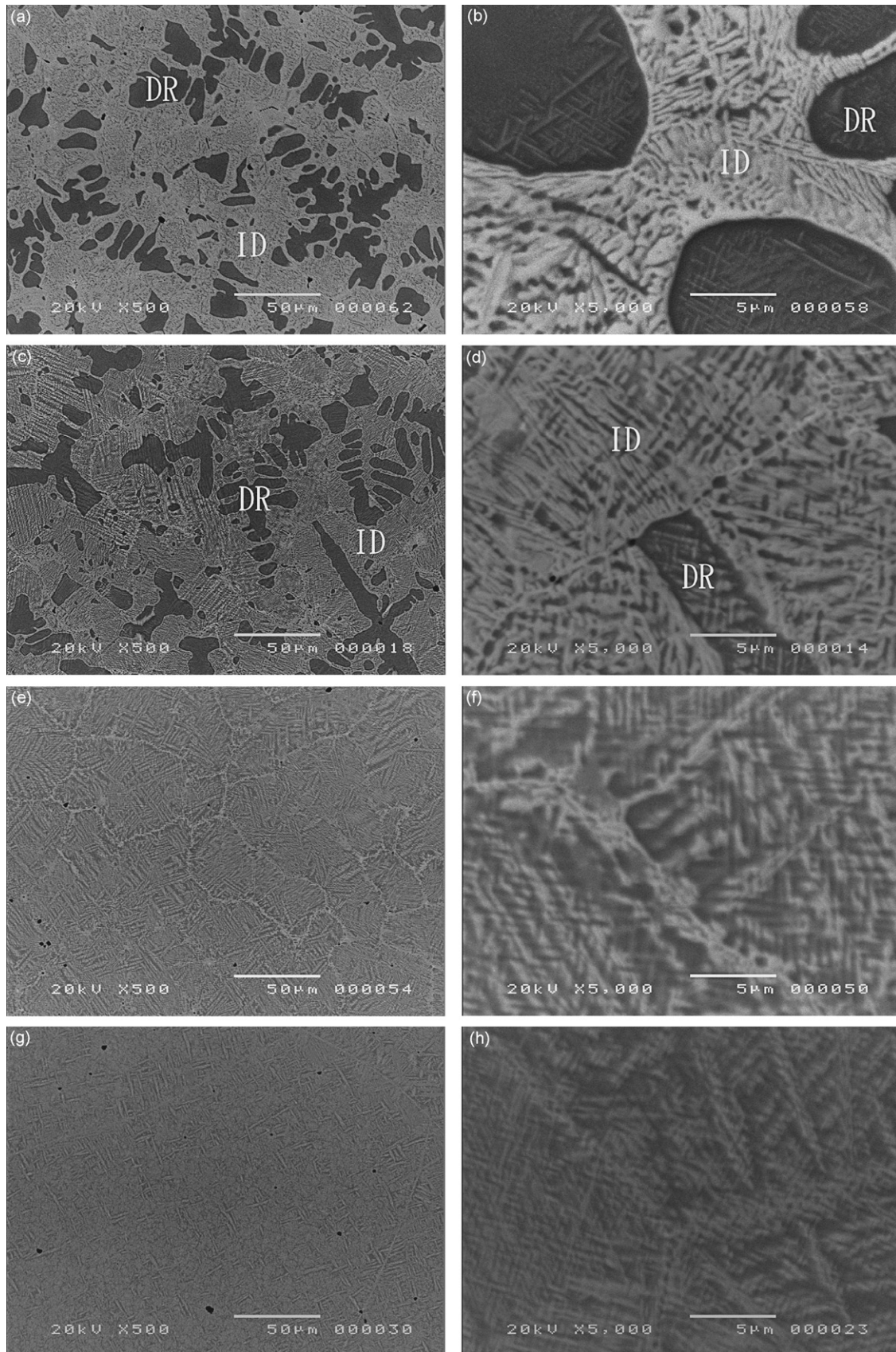


Fig. 3. BEI images of AlCoCrFe_xMo_{0.5}Ni alloys: (a) and (b) for $x=0.6$; (c) and (d) for $x=1.0$; (e) and (f) for $x=1.5$; (g) and (h) for $x=2.0$.

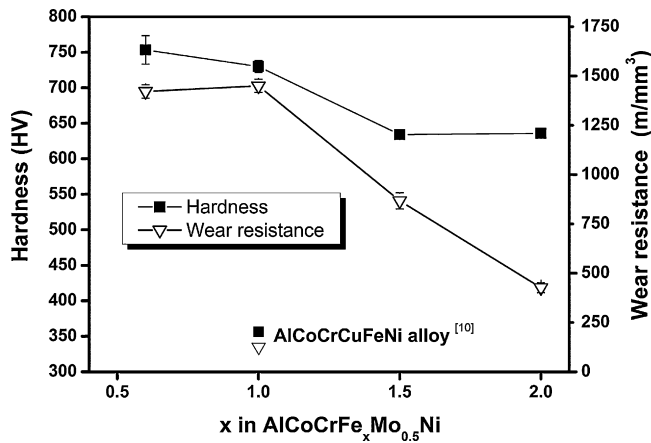


Fig. 4. Plots of Vickers hardness and wear resistance of AlCoCrFe_xMo_{0.5}Ni alloys as functions of Fe content, x.

initially, but precipitate out a large amount of white σ phase during subsequent cooling due to their higher Cr and Mo content. The polygrain morphology of Fe1.5 and Fe2.0 alloys, suggests that their compositions support a small range of melting temperature of the BCC phase. Similarly, BCC-phase grains precipitate out a large amount of σ phase upon cooling. The fact that σ phase precipitates along grain boundaries evidences that the formation of σ phase is a subsequent phase transformation after solidification is completed. The drop in the amount of σ phase or, conversely, the increase in the amount of bcc phase, is attributable to the fact that a higher concentration of Fe effectively reduces the concentrations of constituent Cr and Mo that are required to form σ phase.

3.2. Hardness

Fig. 4 plots the hardness and wear resistance of AlCoCrFe_xMo_{0.5}Ni alloys as a function of Fe content and those of AlCoCrCuFeNi alloy [10] are also shown for comparison. Replacing Cu with a half mole of Mo increases the hardness from Hv 356 to Hv 730, obviously due to the formation of a hard σ phase in place of a soft FCC phase. Furthermore, Fe0.6 and Fe1.0 alloys are about Hv 100 harder than Fe1.5 and Fe2.0 alloys, with an Hv of around

725. This hardness declines as the Fe content increases, because of the corresponding decrease in the amount of σ phase, as analyzed above, since the BCC phase is softer than the σ phase. The hardness of the dendrites (with less σ phase) in the Fe0.6 alloy, and the interdendrites (with more σ phase), were measured to be Hv 643 and Hv 898, respectively.

3.3. Wear behavior

3.3.1. Wear resistance

Fig. 4 also shows the effect of iron content on the wear resistance and the comparison with AlCoCrCuFeNi alloy. The wear resistance is markedly improved by replacing of Cu with a half mole of Mo, which effect is consistent with a large increase in hardness. A comparison of the variation of hardness with Fe content supports Khrushchov's conclusion that the wear resistance of materials is generally proportional to their Vickers' hardness [14]. Notably, however, the wear resistance of Fe2.0 alloy is only half that of Fe1.5 alloy, although they have almost equal hardness. The Fe content that maximizes wear resistance is $x = 0.6$ or 1.0, the main reason for which will be discussed in the following section.

3.3.2. Worn surface and debris

Fig. 5 presents the worn surfaces and debris of the AlCoCrFe_xMo_{0.5}Ni alloys. All alloys reveal their microstructural contrast accompanied with abrasive scratches on the worn surfaces. In Section 3.2, it has been demonstrated that the hardness value of interdendrites is higher than that of dendrites as a result of having more σ phase. Therefore, the σ phase protrudes above BCC phase revealing the microstructures of these HEAs.

Table 2 presents the compositions of the worn surface and debris. In view of the fact that oxygen is in trace amount on all worn surfaces, and large standard deviations of oxygen content are observed for both particle and plate debris, oxidation wear mechanism is excluded and thus the abrasion action by hard oxide particles is the dominant mechanism. Fig. 6 shows the schematic diagram illustrating the coexistence of particle and plate debris during sliding wear and the evidence of those oxidized debris attached on the worn surface. In the wear process, fresh particles and plates are generated from the surface by the plough of hard oxide particles as the width of scratch is of the same order with the size of particle debris. Then, they begin to oxidize under the high interface temperature, which is enhanced by severe defor-

Table 2
Chemical compositions of worn surface and wear debris of AlCoCrFe_xMo_{0.5}Ni alloys in atomic percentage.

x-Value	Al	Cr	Fe	Co	Ni	Mo	O
0.6							
Nominal	19.6	19.6	11.8	19.6	19.6	9.8	
Worn surface	20.2 ± 0.2	18.9 ± 0.2	12.2 ± 0.5	19.1 ± 0.9	19.2 ± 1.0	10.4 ± 0.4	Trace
Worn debris (plate)	8.2 ± 2.6	8.9 ± 3.8	10.8 ± 6.7	8.7 ± 4.2	8.5 ± 4.0	3.8 ± 0.8	51.1 ± 16.1
Worn debris (particle)	7.9 ± 2.2	7.6 ± 3.6	7.2 ± 7.0	7.0 ± 3.3	6.9 ± 3.0	3.5 ± 0.8	59.9 ± 16.8
1.0							
Nominal	18.2	18.2	18.2	18.2	18.2	9.1	
Worn surface	17.5 ± 0.7	18.5 ± 0.7	18.7 ± 0.8	17.7 ± 0.9	18.3 ± 0.8	9.4 ± 0.3	Trace
Worn debris (plate)	9.3 ± 2.3	7.6 ± 1.5	10.7 ± 7.2	7.4 ± 1.6	7.2 ± 1.6	4.3 ± 0.9	53.5 ± 8.5
Worn debris (particle)	6.8 ± 1.5	7.2 ± 2.8	11.7 ± 4.7	6.4 ± 2.9	6.6 ± 2.9	3.6 ± 0.8	57.7 ± 10.9
1.5							
Nominal	16.7	16.7	25	16.7	16.7	8.3	
Worn surface	16.4 ± 1.0	16.7 ± 0.7	25.1 ± 1.1	15.7 ± 1.1	15.7 ± 0.6	10.4 ± 0.3	Trace
Worn debris (plate)	8.2 ± 2.3	10.1 ± 2.0	14.9 ± 2.5	9.5 ± 2.0	9.3 ± 2.2	4.6 ± 0.5	43.4 ± 8.1
Worn debris (particle)	8.6 ± 2.3	7.5 ± 2.5	11.9 ± 3.9	7.2 ± 2.8	7.0 ± 2.5	3.8 ± 0.7	54.0 ± 11.2
2.0							
Nominal	15.4	15.4	30.8	15.4	15.4	7.7	
Worn surface	14.8 ± 0.1	15.3 ± 0.3	30.9 ± 0.7	14.9 ± 0.5	15.0 ± 0.2	9.2 ± 0.3	Trace
Worn debris (plate)	9.9 ± 4.8	9.5 ± 2.5	18.9 ± 6.1	9.3 ± 3.0	9.0 ± 2.2	4.0 ± 0.7	39.4 ± 11.3
Worn debris (particle)	8.2 ± 3.5	6.6 ± 2.5	13.1 ± 4.5	6.6 ± 2.6	6.5 ± 2.5	3.1 ± 0.7	55.9 ± 12.6

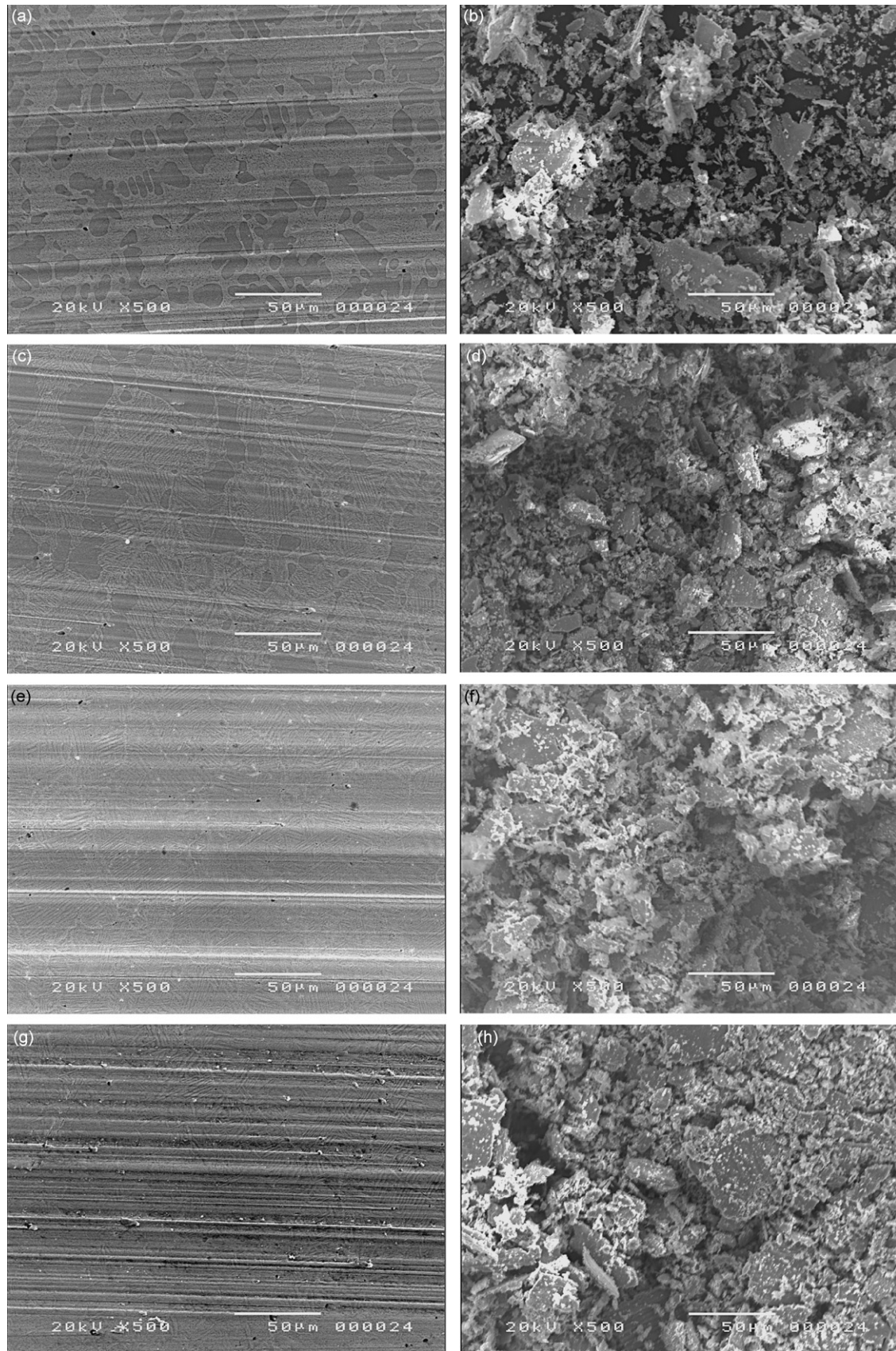


Fig. 5. Microstructures of worn surface and worn debris of AlCoCrFe_xMo_{0.5}Ni alloys with various iron contents: (a) $x=0.6$, worn surface; (b) $x=0.6$, worn debris; (c) $x=1.0$, worn surface; (d) $x=1.0$, worn debris; (e) $x=1.5$, worn surface; (f) $x=1.5$, worn debris; (g) $x=2.0$, worn surface; and (h) $x=2.0$, worn debris.

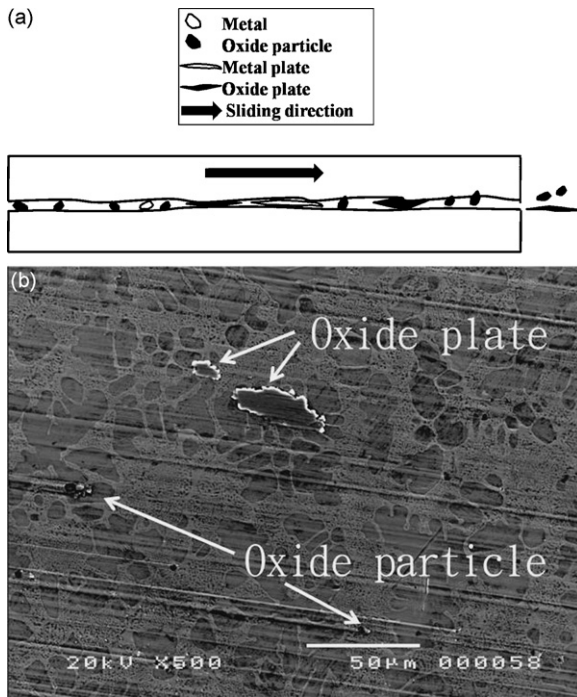


Fig. 6. (a) Schematic diagram illustrating the coexistence of particle and plate debris during sliding wear for the present alloys. (b) The oxide particles and plates observed on the worn surface of Fe0.6 alloy.

mation in the three-body grinding process. It is expected that the plates might be broken into small particles before they leave from the wear trails because of their larger aspect ratio. This is the second source of particle debris. As the particles have a longer time

Table 3

Weight gain of $\text{AlCoCrFe}_x\text{Mo}_{0.5}\text{Ni}$ ($x = 1.5, 2.0$) at 500°C for 24 h.

Alloy	Initial weight (mg)	Final weight (mg)	Weight gain (mg/cm^2)
Fe1.5	1336.33	1336.36	0.0075
Fe2.0	1319.73	1319.84	0.0275

to stay in the interface, particle debris has higher oxygen content than plate debris.

Based on this abrasion mechanism, the reason why Fe2.0 alloy has the similar hardness to Fe1.5 alloy but a significantly lower wear resistance than Fe1.5 alloy is attributable to its much higher oxidation rate which means more oxide particles will form between the surfaces of two counterparts. This explanation is supported by a complementary oxidation test in a furnace at the pin/disk interface flash temperature, 500°C , for 24 h, since the oxidation or weight gain rate of the Fe2.0, indeed markedly exceeds that of the Fe1.5 alloy, as presented in Table 3. In addition, this is supported by the lower oxygen content of plate debris for Fe2.0 alloy as compared with Fe1.5 alloy since the oxide particles of Fe2.0 alloy are much more than that of Fe1.5 alloy and thus plate debris are produced and expelled from the wear trails more quickly.

3.3.3. Coefficient of friction

Fig. 7 plots the variation of the coefficient of friction with the sliding distance of the $\text{AlCoCrFe}_x\text{Mo}_{0.5}\text{Ni}$ alloys. These curves have two typical regimes—one with the peak value μ_p and the other with the steady-state value μ_{ss} , as reported in the literature [15–19]. The steady-state values μ_{ss} of the four alloys are similar, about 0.74, but the fluctuation vary negatively with the iron content. Fig. 3 shows that the Fe0.6 and Fe1.0 alloys have the dendritic microstructure whereas the Fe1.5 and Fe2.0 alloy have the polygrain microstructure. Since the interdendrites of Fe0.6 and Fe1.0 alloys contain more

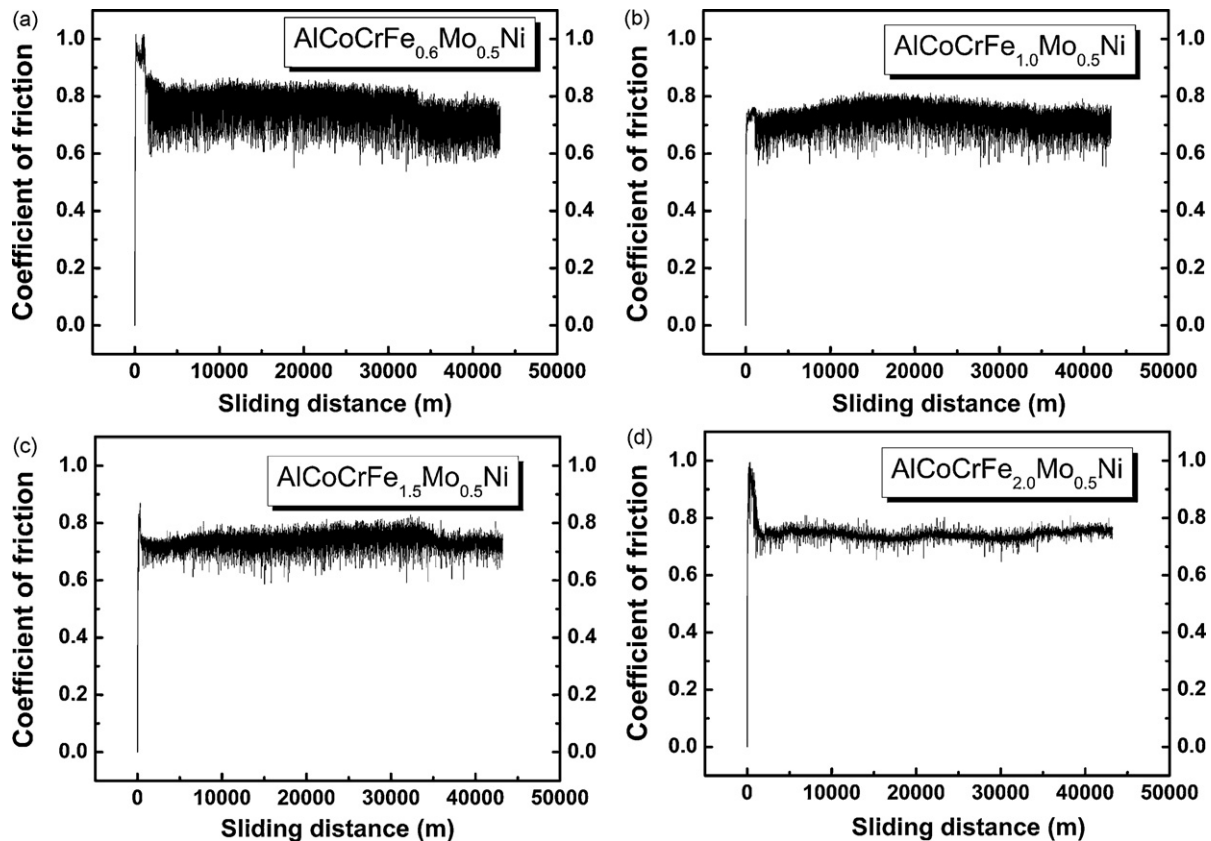


Fig. 7. Coefficient of friction vs. sliding distance curves for $\text{AlCoCrFe}_x\text{Mo}_{0.5}\text{Ni}$ alloys: (a) $x = 0.6$, (b) $x = 1.0$, (c) $x = 1.5$, and (d) $x = 2.0$.

hard σ phase than the dendrites, the former protrude above the latter resulting larger surface roughness. This thus indicates that the uniformity of microstructure will influence the fluctuation. Fe2.0 alloy has the most uniform microstructure, and therefore exhibits the smallest fluctuation, while Fe0.6 alloy exhibits the largest fluctuation because of its apparent dendritic structure.

4. Conclusions

The new high-entropy alloy system, AlCoCrFe_xMo_{0.5}Ni, derived from the AlCoCrCuFeNi system by replacing Cu with Mo has a markedly higher hardness and wear resistance because of the formation of a hard σ phase rather than a soft FCC phase. As the iron content increases from Fe0.6 and Fe1.0 alloy to Fe1.5 and Fe2.0, the microstructure changes from “dendritic” to “polygrain”, and the relative volume fraction of BCC to σ phase increases. Both the hardness and the wear resistance drop as the Fe content increases since the σ phase is harder than the BCC phase. However, the wear resistance of the Fe2.0 alloy is much less than that of Fe1.5, because the oxidation rate of Fe2.0 alloy markedly exceeds that of Fe1.5 alloy. The composition analysis of the worn surface and worn debris indicates that the major wear mechanism of all these HE alloys is abrasion. A lower Fe content, such as $x=0.6$ or 1.0 , is preferred over $x=1.5$ and 2.0 if high hardness, excellent wear resistance and oxidation resistance are required.

Acknowledgement

The authors are pleased to acknowledge the financial support for this research by Ministry of Economic Affairs R.O.C. under Grant No. 97-EC-17-A-08-S1-03.

References

- [1] J.W. Yeh, S.K. Chen, S.J. Lin, J.Y. Gan, T.S. Chin, T.T. Shun, C.H. Tsau, S.Y. Chang, Nanostructured high-entropy alloys with multiple principal elements: novel alloy design concepts and outcomes, *Adv. Eng. Mater.* 6 (2004) 299–303.
- [2] J.W. Yeh, Recent progress in high-entropy alloys, *Ann. Chim.: Sci. Mat.* 31 (2006) 633–648.
- [3] C.J. Tong, Y.L. Chen, S.K. Chen, J.W. Yeh, T.T. Shun, C.H. Tsau, S.J. Lin, S.Y. Chang, Microstructure characterization of Al_xCoCrCuFeNi high-entropy alloy system with multiprincipal elements, *Metall. Mater. Trans. A* 36A (2005) 881–893.
- [4] C.Y. Hsu, J.W. Yeh, S.K. Chen, T.T. Shun, Wear resistance and high-temperature compression strength of Fcc CuCoNiCrAl_{0.5}Fe alloy with boron addition, *Metall. Mater. Trans. A* 35A (2004) 1465–1469.
- [5] M.R. Chen, S.J. Lin, J.W. Yeh, S.K. Chen, Y.S. Huang, C.P. Tu, Microstructure and properties of Al_{0.5}CoCrCuFeNiTi_x ($x=0-2.0$) high-entropy alloys, *Mater. Trans.* 47 (2006) 1395–1401.
- [6] M.R. Chen, S.J. Lin, J.W. Yeh, S.K. Chen, Y.S. Huang, M.H. Chuang, Effect of vanadium addition on the microstructure, hardness, and wear resistance of Al_{0.5}CoCrCuFeNi high-entropy alloy, *Metall. Mater. Trans. A* 37A (2006) 1363–1369.
- [7] X.F. Wang, Y. Zhang, Y. Qiao, G.L. Chen, Novel microstructure and properties of multicomponent CoCrCuFeNiTi_x alloys, *Intermetallics* 15 (2007) 357–362.
- [8] Y.J. Zhou, Y. Zhang, Y.L. Wang, G.L. Chen, Solid solution alloys of AlCoCrFeNiTi_x with excellent room-temperature mechanical properties, *Appl. Phys. Lett.* 90 (2007) 181904.
- [9] C.J. Tong, M.R. Chen, S.K. Chen, J.W. Yeh, T.T. Shun, S.J. Lin, S.Y. Chang, Mechanical performance of the Al_xCoCrCuFeNi high-entropy alloy system with multiprincipal elements, *Metall. Mater. Trans. A* 36A (2005) 1263–1271.
- [10] J.M. Wu, S.J. Lin, J.W. Yeh, S.K. Chen, Y.S. Huang, Adhesive wear behavior of Al_xCoCrCuFeNi high-entropy alloys as a function of aluminum content, *Wear* 261 (2006) 513–519.
- [11] C.P. Lee, Y.Y. Chen, C.Y. Hsu, J.W. Yeh, H.C. Shih, The effect of boron on the corrosion resistance of the high entropy alloys Al_{0.5}CoCrCuFeNiB_x, *J. Electrochem. Soc.* 154 (2007) C424–C430.
- [12] M.F. Ashby, J. Abulawi, H.S. Kong, Temperature maps for frictional heating in dry sliding, *Tribol. Trans.* 34 (1991) 577–587.
- [13] H. Baker (Ed.), *Metals Handbook*, vol. 3, 10th ed., ASM International, Metals Park, OH, 1992.
- [14] M.M. Khrushchov, Principles of abrasive wear, *Wear* 28 (1974) 69–88.
- [15] P.J. Blau, *Friction and Wear Transition of Materials*, Noyes Publishing, New Jersey, 1989.
- [16] N.P. Suh, *Tribophysics*, Prentice-Hall, New Jersey, 1986.
- [17] P.J. Blau, Model for run-in and other transitions in sliding friction, *J. Tribol.* 109 (1987) 537–544.
- [18] D.A. Rigney, J.P. Hirth, Plastic deformation and sliding friction of metals, *Wear* 53 (1979) 345–370.
- [19] Z.N. Farhat, Contribution of crystallographic texturing to the sliding friction behaviour of FCC and HCP metals, *Wear* 250 (2001) 401–408.